Department: Computer Science & Engineering

2nd Year 2nd Semester Examination-2020 (Session 2018-19)

Subject: Electrical Devices and Instrumentation

Subject code: EEE-407

Marks: 70

Time: 3.00 Hours

[Answer any five questions. Assume reasonable value for any missing data.]

1. a) What is Transformer? Why frequency of primary and secondary of a transformer are same? 5
   b) Why Transformer rated at KVA ? 2
   c) Explain Transformer name plate Data. 3
   d) Find the all day efficiency of a 50 KVA distribution transformer having full load efficiency of 94% and full load copper loss equal to constant Iron losses. The loading p.f 1.0 is as (i) No load at 10 hours; (ii) half load at 5 hours; (iii) 25% load at 6 hours; (iv) full load at 3 hours. 4

2. a) Why do we need a starter to start a DC motor? 3
   b) How can we control the speed of stepper motor? 3
   c) Write about Permanent magnet Brushless AC Generator with circuit Diagram. 4
   d) A 3 phase induction motor is wound for 4 poles and is supplied from 50-Hz system. Calculate (i) the synchronous speed (ii) The rotor speed when slip is 4% and (iii) Rotor frequency when rotor runs at 600 rpm. 4

3. a) What are the desirable characteristics of a transducer elements? 4
   b) State the difference between analog and digital indicators. 3
   c) Explain the operating principle of a ramp type DVM. 4
   d) A capacitive transducer consists of two plates of diameter 2 cm each, separated by an air gap of 0.25 mm. Find the displacement sensitivity. 3

4. a) Classify DC generator. 3
   b) What is back e.m.f ? Write down the significance of back e.m.f. 3
   c) A 25-kW, 250-V d.c shunt generator has armature and field resistance of 0.06Ω and 100 Ω respectively. Determine the total armature power developed when working (i) as a generator delivering 25kW output and (ii) As a motor taking 25kW input. 5
   d) Why single phase induction motor is not self-starting? 3

5. a) Classify digital voltmeters. 2
   b) What is instrumentation amplifier? Write down the key features of instrumentation amplifier. 4
   c) Explain the construction and operation of a LED display. 6
   d) Compare LED and LCD displays. 2

6. a) What are the key features of instrumentation amplifiers? 2
   b) Explain the construction and operation of a Piezo-electric transducer. 4
   c) Explain the working principle of a Thermo-electric generator. 5
   d) Discuss about functional elements of a measurements system. 3

X a) What do you understand by DAS? 2
   b) Discuss the operation of a large-scale DAS with required block diagram. 6
   c) Show the schematic block diagram of a generalized data acquisition system. 4
   d) State the objectives of a DAS. 2